

Q1.

Write a function `check_even_odd(n)` that accepts a number and prints whether the number is even or odd.

```
def check_even_odd(num):
    if num % 2 == 0:
        print(f"{num} is even")
    else:
        print(f"{num} is odd")

check_even_odd(68)
```

Q2.

Write a function `check_number(n)` that accepts a number and prints whether it is positive, negative, or zero.

```
def check_number(n):
    if n < 0:
        print(f"{n} is a negative num")
    elif n > 0:
        print(f"{n} is a positive num")
    else:
        print(f"{n} is 0")

check_number(20)
```

Q3.

Write a function `find_largest(a, b)` that accepts two numbers and returns the largest number.

```
def find_largest(a, b):
    if a >= b:
        return a
    else:
        return b

print(find_largest(0.9, 1.9))
```

Q4.

Write a function `find_largest(a, b, c)` that finds the largest among three numbers.

```
def find_largest(a, b, c):
    if a >= b and a >= c:
        return a
    elif b >= a and b >= c:
        return b
    else:
        return c

print(find_largest(100, 22, 10))
```

Q5.

Write a function `sum_natural(n)` that calculates the sum of numbers from 1 to n.

```
def sum_natural(n):
    total = 0

    for i in range(1, n + 1):
        total += i

    print(total)

sum_natural(100)
```

Q6.

Write a function `multiplication_table(n)` that prints the multiplication table of a given number from 1 to 10.

```
def multiplication_table(n):
    if n < 1 or n > 10:
        return None

    for i in range(1, 11):
        print(i * n, end=" ")

multiplication_table(1)
```

Q7.

Write a function `factorial(n)` that calculates the factorial of a number.

```
def factorial(n):
    fact = 1

    for i in range(1, n + 1):
        fact *= i

    return fact

print(factorial(5))
```

Q8.

Write a function `count_digits(n)` that counts the number of digits in a number.

```
def count_digits(n):
    n = abs(n)

    if n == 0:
        return 1

    count = 0

    while n > 0:
        count += 1
        n //= 10

    return count

print(count_digits(12345))
```

Q9.

Write a function `reverse_number(n)` that reverses a given number.

```
def reverse_number(n):
    reverse = 0

    while n > 0:
        digit = n % 10
        reverse = reverse * 10 + digit
        n //= 10

    return reverse

print(reverse_number(12345))
```

Q10.

Write a function `check_prime(n)` that checks whether a number is prime.

```
def check_prime(n):
    if n < 2:
        return False
```

```
    for i in range(2, n):
        if n % i == 0:
            return False

    return True

print(check_prime(29))
```

Q11.

Write a function `count_characters(text)` that counts the number of characters in a string.

```
def count_characters(text):
    count = 0

    for char in text:
        count += 1

    return count

print(count_characters("Python"))
```

Q12.

Write a function `count_vowels(text)` that counts the number of vowels in a string.

```
def count_vowels(text):
    count = 0

    for char in text.lower():
        if char in "aeiou":
            count += 1

    return count

print(count_vowels("education"))
```

Q13.

Write a function `count_consonants(text)` that counts the number of consonants in a string.

```
def count_consonants(text):
    count = 0

    for char in text.lower():
        if char.isalpha() and char not in "aeiou":
            count += 1

    return count

print(count_consonants("Python Programming"))
```

Q14.

Write a function `count_vowels_consonants(text)` that separately counts vowels and consonants.

```
def count_vowels_consonants(text):
```

```
vowels = 0
consonants = 0

for char in text.lower():
    if char in "aeiou":
        vowels += 1
    elif char.isalpha():
        consonants += 1

print("Vowels :", vowels)
print("Consonants :", consonants)

count_vowels_consonants("Python Programming")
```

Q15.

Write a function `reverse_string(text)` that reverses a string.

```
def reverse_string(text):
    reverse = ""

    for char in text:
        reverse = char + reverse

    print("Using loop :", reverse)
    print("Using slicing :", text[::-1])

reverse_string("Python")
```

Q16.

Write a function `check_palindrome(text)` that checks whether a string is a palindrome.

```
def check_palindrome(text):
    if text == text[::-1]:
        print("Palindrome")
    else:
        print("Not Palindrome")

check_palindrome("madam")
```

Q17.

Write a function `count_words(text)` that counts the number of words in a sentence.

```
def count_words(text):
    count = 0
    in_word = False

    for char in text:
        if char != " " and not in_word:
            count += 1
            in_word = True
        elif char == " ":
            in_word = False

    return count

print(count_words("Python is easy to learn"))
```

Q18.

Write a function `character_frequency(text, ch)` that counts how many times a particular character occurs.

```
def character_frequency(text, ch):
    count = 0

    for char in text:
        if char == ch:
            count += 1

    return count

print(character_frequency("programming", "g"))
```

Q19.

Write a function `remove_spaces(text)` that removes all spaces from a string.

```
def remove_spaces(text):
    result = ""

    for char in text:
        if char != " ":
            result += char
```

```
        return result

print(remove_spaces("Python Programming Language"))
```

Q20.

Write a function `convert_uppercase(text)` that converts a string to uppercase.

```
def convert_uppercase(text):
    print(text.upper())

    result = ""

    for char in text:
        if "a" <= char <= "z":
            result += chr(ord(char) - 32)
        else:
            result += char

    print(result)

convert_uppercase("Python Programming")
```

Q21.

Write a function `count_case(text)` that counts uppercase characters, lowercase characters, digits, and spaces.

```
def count_case(text):
    uppercase = 0
    lowercase = 0
    digits = 0
    spaces = 0

    for char in text:
        if char.isupper():
            uppercase += 1
        elif char.islower():
            lowercase += 1
        elif char.isdigit():
            digits += 1
        elif char == " ":
            spaces += 1

    print("Uppercase :", uppercase)
    print("Lowercase :", lowercase)
    print("Digits :", digits)
    print("Spaces :", spaces)

count_case("Python123 ABC")
```

Q22.

Write a function `first_character(text)` that prints the first character of a string.

```
def first_character(text):
    for char in text:
        print(char)
        break

first_character("PYTHON")
```

Q23.

Write a function `last_character(text)` that prints the last character of a string.

```
def last_character(text):
    print(text[-1])

    last = ""
    for char in text:
        last = char

    print(last)

last_character("PYTHON")
```

Q24.

Write a function `display_characters(text)` that prints every character of a string on a separate line.

```
def display_characters(text):
    for char in text:
        print(char)

display_characters("PYTHON")
```

Q25.

Write a function `display_position(text)` that displays every character along with its position.

```
def display_position(text):
    for i in range(len(text)):
        print("Position", i, ":", text[i])

display_position("JAVA")
```

Q26.

Write a function `remove_vowels(text)` that removes all vowels from a string.

```
def remove_vowels(text):
    result = ""

    for char in text:
        if char.lower() not in "aeiou":
            result += char

    return result
```

```
print(remove_vowels("Python Programming"))
```

Q27.

Write a function `find_longest_word(text)` that finds the longest word in a sentence.

```
def find_longest_word(text):
    longest = ""
    current = ""

    for char in text:
        if char != " ":
            current += char
        else:
            if len(current) > len(longest):
                longest = current
            current = ""

    if len(current) > len(longest):
        longest = current

    return longest

print(find_longest_word("Python Programming Language"))
```

Q28.

Write a function that separately counts the occurrence of a, e, i, o, and u.

```
def count_vowels_separately(text):
    a = e = i = o = u = 0

    for char in text.lower():
        if char == "a":
            a += 1
        elif char == "e":
            e += 1
        elif char == "i":
            i += 1
        elif char == "o":
            o += 1
        elif char == "u":
            u += 1

    print("a =", a)
    print("e =", e)
    print("i =", i)
    print("o =", o)
    print("u =", u)

count_vowels_separately("education")
```

Q29.

Write a function `star_pattern(n)` that prints the star pattern shown in the assignment.

```
def star_pattern(n):
    for i in range(1, n + 1):
        for j in range(i):
            print("*", end="")
```

```
        print()

star_pattern(5)
```

Q30.

Write a function `number_pattern(n)` that prints the number pattern shown in the assignment.

```
def number_pattern(n):
    for i in range(1, n + 1):
        for j in range(1, i + 1):
            print(j, end="")
        print()

number_pattern(5)
```